Bayesian Inference in Cosmology

From Bayes' theorem to Euclid-era likelihoods

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Plan: two hours, two parts

Hour 1 (done)

Min	Part
0–20	I. Foundations: Bayes' theorem, priors, χ^2
20–38	II. Hubble's law; Fisher and Hessian; Metropolis–Hastings
38–43	Break
43–51	Live: the notebook (grid, Fisher, MCMC)
51–60	Posterior three ways; Fisher forecasts

Hour 2 (today)

Min	Part
0–4	Recap of hour 1
4–8	Evidence and model selection
8–23	More samplers: proposals, Gibbs, ensemble, HMC, nested
23–28	Break
28–42	Fisher matrix in depth: toolbox, LSS, step sizes, validation, where it fails
42–44	Our own check: Nautilus vs Fisher
44–53	Live: cosmicfishpie
53–56	cloelib and cloelike
56–60	Exercises, take-home

Sources: Padilla et al., arXiv:1903.11127; Verde, arXiv:0911.3105; Euclid papers arXiv:2303.09451 (MontePython validation) and arXiv:2405.06047 (neutrino sector).

Hour 1 used **mock data**; hour 2 uses Euclid-like forecast specifications and published results.

We have one Universe

Frequentist

- Probability = long-run frequency of outcomes
- Parameters are fixed, data are random
- Needs repeatable experiments
- No prior: parameters are not random, so they get no probability; earlier knowledge enters as extra data or fixed constraints

Bayesian

- Probability = degree of belief
- Data are fixed (observed); parameters carry a distribution
- Works for a single realisation
- Prior $p(\boldsymbol{\theta} \mid M)$: what we knew before the data; it must be stated, and enough data overwhelm it

The question we actually want to answer

How probable are the parameters, given the data we have? $p(\boldsymbol{\theta} \mid D)$

Both camps use the same rules of probability; they differ in what is allowed to be a probability.

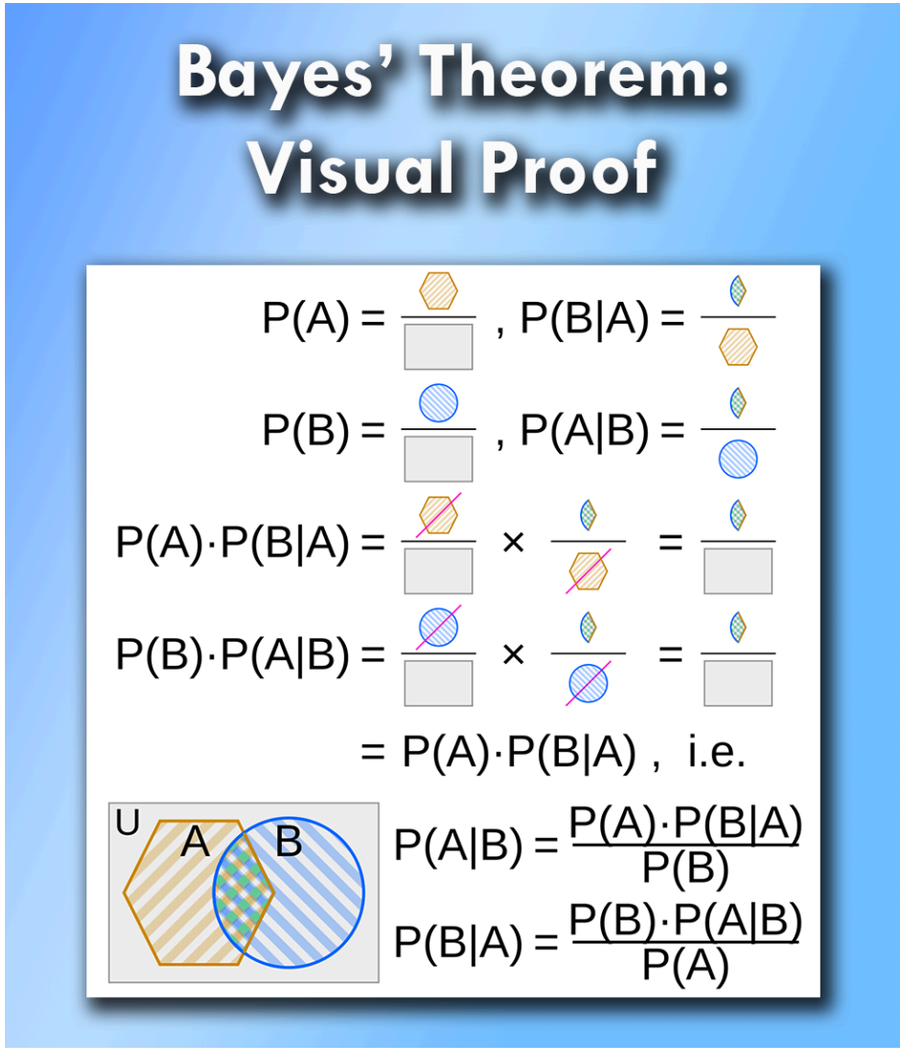
Bayes' theorem

$$p(\boldsymbol{\theta} \mid D, M) = \frac{p(D \mid \boldsymbol{\theta}, M)\; p(\boldsymbol{\theta} \mid M)}{p(D \mid M)}$$

- **Posterior**: what we know after seeing the data
- **Likelihood**: forward model; the data we expect for given parameters
- **Prior**: what we knew before (earlier experiments, physical limits)
- **Evidence**: $Z = p(D|M) = \int \mathrm{d}\boldsymbol{\theta}\, p(D|\boldsymbol{\theta}, M)\, p(\boldsymbol{\theta}|M)$

For parameter estimation the evidence is only a normalisation; it becomes the main object when we compare models (Part III).

Bayes' theorem is the product rule, read twice



Count the overlap $A \cap B$ in two ways:

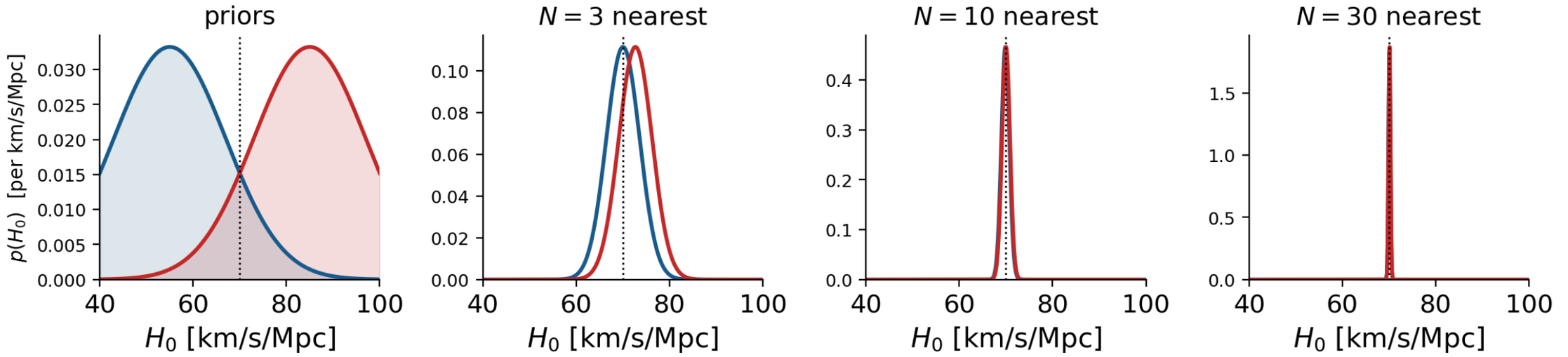
$$p(A, B) = p(A \mid B)\, p(B) = p(B \mid A)\, p(A)$$

$$\Rightarrow \quad p(A \mid B) = \frac{p(B \mid A)\, p(A)}{p(B)}$$

- The theorem itself is one line of algebra.
- The content is in using it for parameters: $A \to \boldsymbol{\theta}$, $B \to D$.
- The hard parts are the likelihood, the prior and the integrals.

Figure: adapted from "Bayes theorem visual proof" by Cmglee, Wikimedia Commons, CC BY-SA 4.0 (en.wikipedia.org/wiki/Bayes%27_theorem); title and background added by statisticsglobe.com.

Priors are overwhelmed by data; updating is consistent



Two observers with Gaussian priors on H_0: N(55, 12) (blue) and N(85, 12) (red). Same mock data: the 3, 10, 30 nearest galaxies. One parameter only: the offset a is fixed to 0 here. Each panel has its own y-scale.

$$\frac{1}{\sigma_N^2} = \frac{1}{\sigma_0^2} + \sum_{i=1}^{N} \frac{d_i^2}{\sigma^2}, \qquad \mu_N = \sigma_N^2 \left(\frac{\mu_0}{\sigma_0^2} + \sum_{i=1}^{N} \frac{d_i \, cz_i}{\sigma^2} \right)$$

- The prior term $1/\sigma_0^2$ is fixed; the data term grows with N, so both observers converge
- The sums simply add: updating galaxy by galaxy (posterior of step 1 = prior of step 2) gives the same result as analysing all data jointly
- If the prior still matters, the data are not informative enough

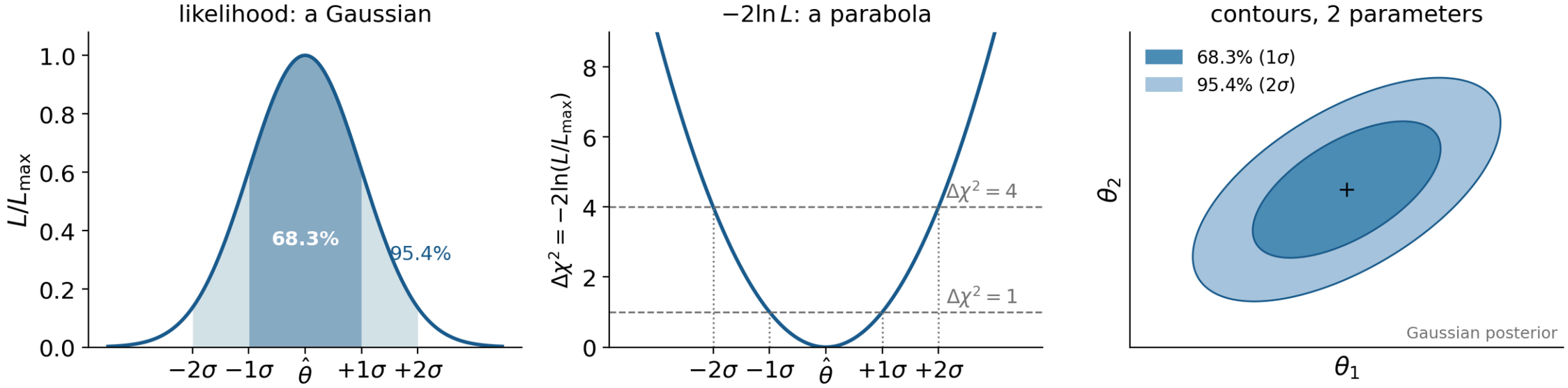
Gaussian likelihood and χ² (“chi-squared”)

$$L(\boldsymbol{\theta}) \propto \exp\left[-\tfrac{1}{2}\sum_{i=1}^{N}\frac{\left(d_i - t_i(\boldsymbol{\theta})\right)^2}{\sigma_i^2}\right]$$

$$\Rightarrow \quad -2\ln L = \chi^2(\boldsymbol{\theta}) + \text{const}, \qquad \chi^2 = \sum_{i=1}^{N}\frac{\left(d_i - t_i(\boldsymbol{\theta})\right)^2}{\sigma_i^2}$$

- d_i: measured data point (here: cz of galaxy i)
- $t_i(\boldsymbol{\theta})$: theory prediction for parameters $\boldsymbol{\theta}$ (here: $a + H_0 d_i$)
- σ_i: error of d_i

χ² (chi-squared): sum of squared, error-weighted residuals. Minimum χ² = best fit; $\Delta\chi^2 = \chi^2 - \chi^2_{\min}$ sets the error bars.



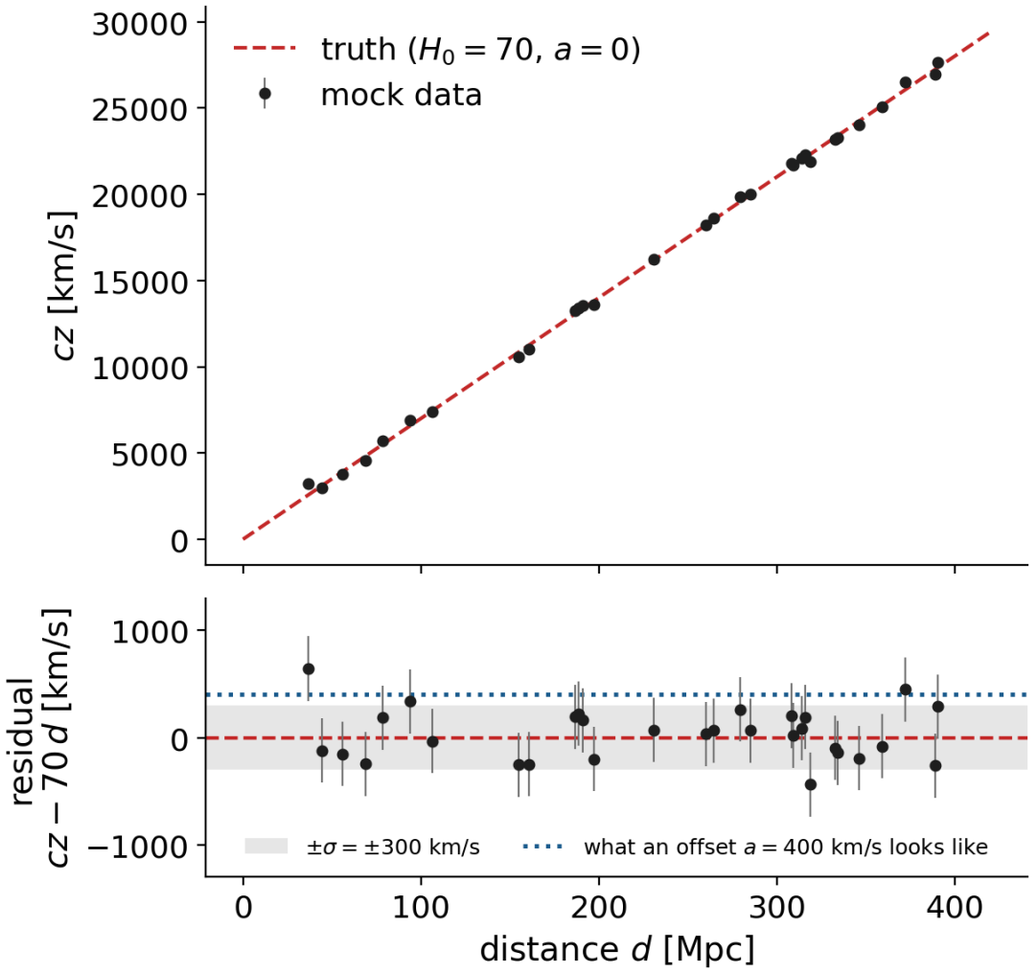
A **contour** is a line of constant posterior; the region inside holds a fixed probability: 1σ = 68.3 %, 2σ = 95.4 % (Gaussian posterior).

Worked example: Hubble's law with a nuisance offset

$$cz_i = a + H_0\, d_i + \epsilon_i$$

Gaussian noise ϵ_i with $\sigma = 300$ km/s (peculiar velocities).

- Data: velocities cz_i at distances d_i. As on the χ^2 slide: data cz_i, theory $t_i(\boldsymbol{\theta}) = a + H_0 d_i$ (here d_i is a distance)
- Parameters $\boldsymbol{\theta} = (H_0, a)$; $\chi^2 = \sum_i (cz_i - a - H_0 d_i)^2/\sigma^2$
- **Nuisance parameter**: a parameter the model needs, but we are not interested in its value. Here a: a zero-point offset or bulk flow
- Flat priors: $H_0 \in [40, 100]$ km/s/Mpc, $a \in [-500, 500]$ km/s
- $N = 30$ mock galaxies, $d \in [20, 400]$ Mpc; truth $H_0 = 70$ km/s/Mpc, $a = 0$
- a and H_0 are degenerate: the line pivots about the centre of the data, so a larger a needs a smaller H_0



Next: we analyse this with the **Fisher matrix** and with **MCMC**, and compare both with the exact grid posterior.

Fisher matrix and Hessian: the Gaussian approximation

Expand $\ln L$ around its maximum $\hat{\boldsymbol{\theta}}$ (the gradient vanishes there):

$$\ln L(\boldsymbol{\theta}) \approx \ln L(\hat{\boldsymbol{\theta}}) - \tfrac{1}{2} \sum_{i,j} (\theta_i - \hat{\theta}_i)\, H_{ij}\, (\theta_j - \hat{\theta}_j)$$

$$H_{ij} = - \left. \frac{\partial^2 \ln L}{\partial\theta_i\, \partial\theta_j} \right|_{\hat{\boldsymbol{\theta}}}$$

Hessian: the curvature of $\ln L$ at the peak. Sharp peak = large curvature = small errors. With flat priors the posterior is a Gaussian with covariance H^{-1}.

$$F_{ij} = \left\langle - \frac{\partial^2 \ln L}{\partial\theta_i\, \partial\theta_j} \right\rangle = \sum_k \frac{1}{\sigma_k^2} \frac{\partial t_k}{\partial\theta_i} \frac{\partial t_k}{\partial\theta_j}$$

- **Fisher matrix** F: the Hessian averaged over data realisations. It needs only the model $t(\boldsymbol{\theta})$ and the errors, no data
- Marginal error $\sigma(\theta_i) = \sqrt{(F^{-1})_{ii}}$; conditional error $1/\sqrt{F_{ii}}$ (all others fixed)
- Our toy: $\partial t_i/\partial a = 1$, $\partial t_i/\partial H_0 = d_i$:

$$F = \frac{1}{\sigma^2} \begin{pmatrix} N & \sum_i d_i \\ \sum_i d_i & \sum_i d_i^2 \end{pmatrix}$$

Linear model + Gaussian noise: Fisher is exact
t is linear in $\boldsymbol{\theta}$, so $\ln L$ is exactly quadratic: H depends neither on $\boldsymbol{\theta}$ nor on the data, $H = F$, and the posterior is exactly Gaussian (inside the flat priors).
Why? Differentiate χ^2 twice: exercise.

Next: sampling the posterior with MCMC; then grid, Fisher and MCMC live in the notebook.

Sampling: Metropolis–Hastings

1. Start at $\boldsymbol{\theta}_i$; compute $p_i \propto \mathcal{L}\,\pi$
2. Propose $\boldsymbol{\theta}_c \sim q(\boldsymbol{\theta}_c|\boldsymbol{\theta}_i)$
3. Accept with probability $\min[1, p_c/p_i]$ (symmetric q)
4. If rejected, repeat $\boldsymbol{\theta}_i$ in the chain
5. Iterate; discard burn-in

Why this samples the posterior

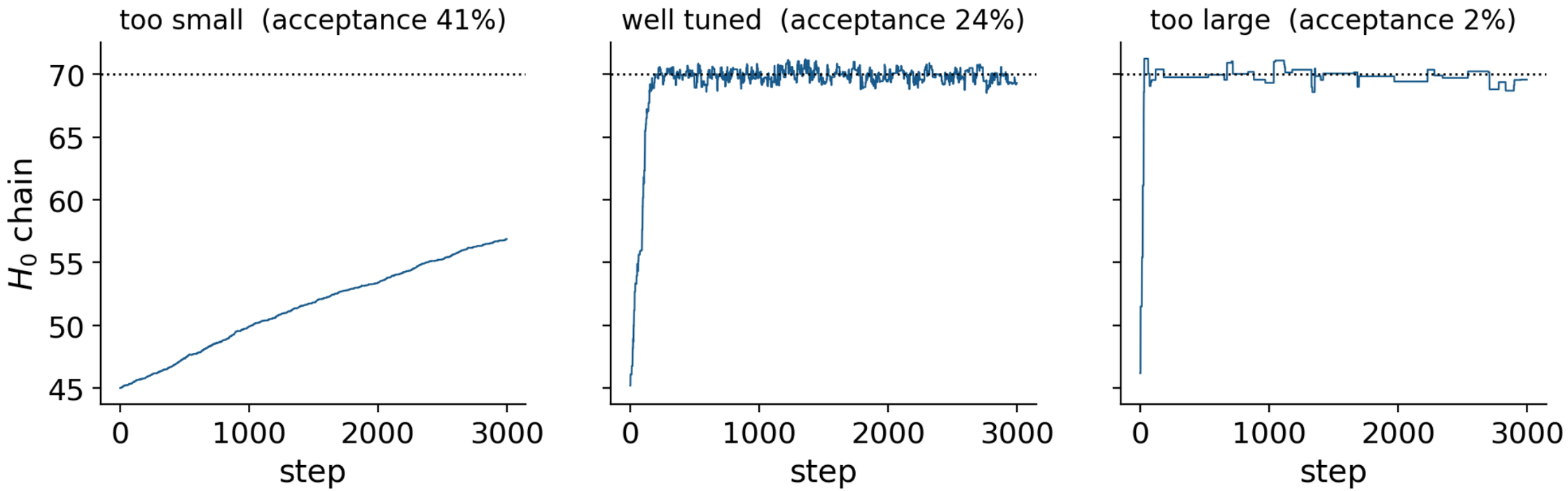
- Only the ratio p_c/p_i enters, so the unknown evidence cancels
- The chain is a Markov process whose stationary distribution is the posterior

Chain density $\rightarrow$ posterior; $\langle f \rangle \approx \frac{1}{N}\sum_i f(\boldsymbol{\theta}_i)$.

Convergence: Gelman–Rubin $\hat{R} \rightarrow 1$. Here (4 chains, burn-in 1000): $\hat{R} = 1.003$, acceptance about 22%.

Cost scales roughly linearly with the number of parameters; a grid scales exponentially.

Metropolis–Hastings: the proposal width matters



Same posterior, same start (H_0=45), 3000 steps each. Left: too small, the chain barely moves. Middle: well tuned. Right: too large, almost everything is rejected. A target acceptance of 20–30% is typical for a few parameters.

Always check: several chains from dispersed starts, $\hat{R}$, autocorrelation, burn-in.

Live: run it in the notebook

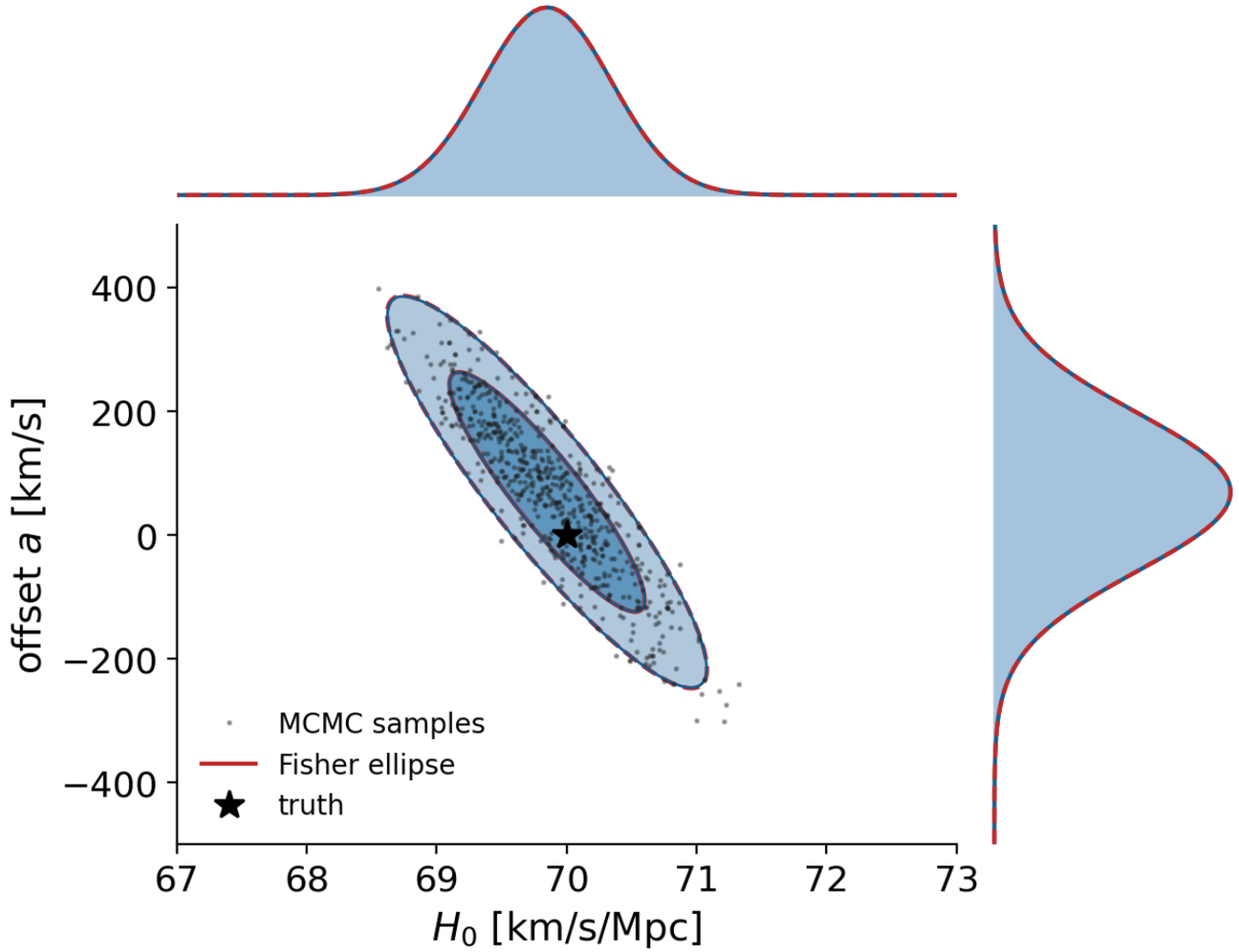
Switch to **bayes_toy_examples.ipynb**: same seed and same mock data as on these slides.

1. Cell 1: generate the 30 mock galaxies (truth $H_0 = 70$ km/s/Mpc, $a = 0$)
2. Section 2: χ^2 on a grid in (H_0, a), the posterior and its 68 % / 95 % contours
3. Section 3: the Fisher matrix $F = X^T X / \sigma^2$, marginal vs conditional errors
4. Section 4: Metropolis–Hastings, 4 chains, acceptance and R-hat

Before running, ask:
Which parameter is better constrained? Are H_0 and a correlated or anti-correlated? Will the Fisher ellipse match the grid?

If the live run fails: the next slide shows the same results.

The posterior, three ways



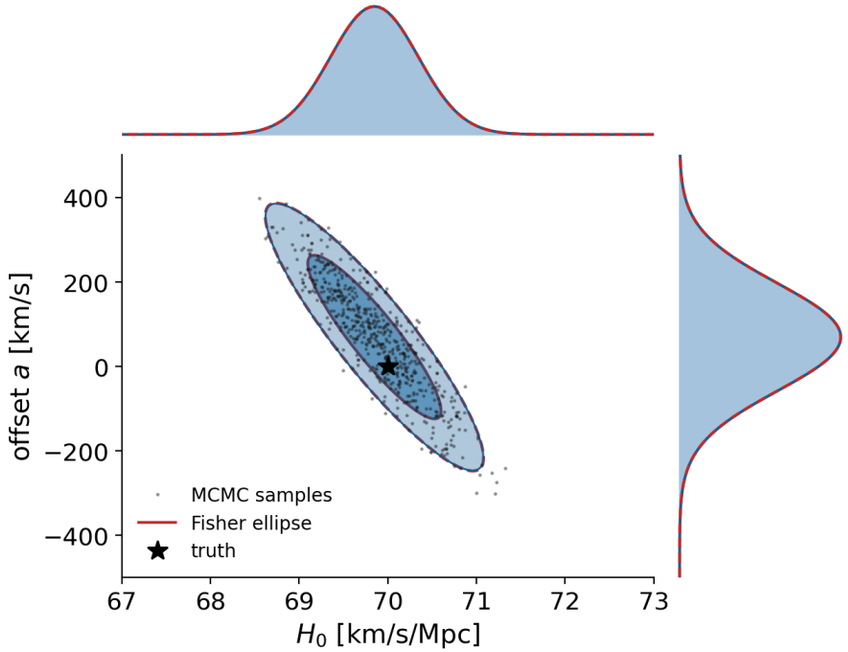
Results (km/s, km/s/Mpc)

Method	Result
Fisher	$H_0 = 69.8 \pm 0.5$
Fisher	$a = 69 \pm 128$
MCMC	$H_0 = 69.8 \pm 0.5$
MCMC	$a = 71 \pm 124$

- Shaded: grid posterior (68%, 95%)
- Dots: Metropolis samples
- Red: Fisher ellipse and Gaussian marginals
- Linear model and Gaussian noise: the three agree

Recap of hour 1

- Posterior = likelihood × prior / evidence: $p(\boldsymbol{\theta}|D) \propto p(D|\boldsymbol{\theta})\, p(\boldsymbol{\theta})$
- Gaussian likelihood: $-2 \ln L = \chi^2$; joint regions from $\Delta\chi^2$ (2.30 and 6.17 for two parameters)
- Gaussian approximation: the Hessian of $-\ln L$ at the peak gives the covariance; quote the marginal error $\sqrt{(H^{-1})_{ii}}$, not $1/\sqrt{H_{ii}}$
- Metropolis–Hastings: only posterior ratios enter, so the evidence cancels
- Check convergence: several chains, $\hat{R} \to 1$, acceptance 20–30%, burn-in

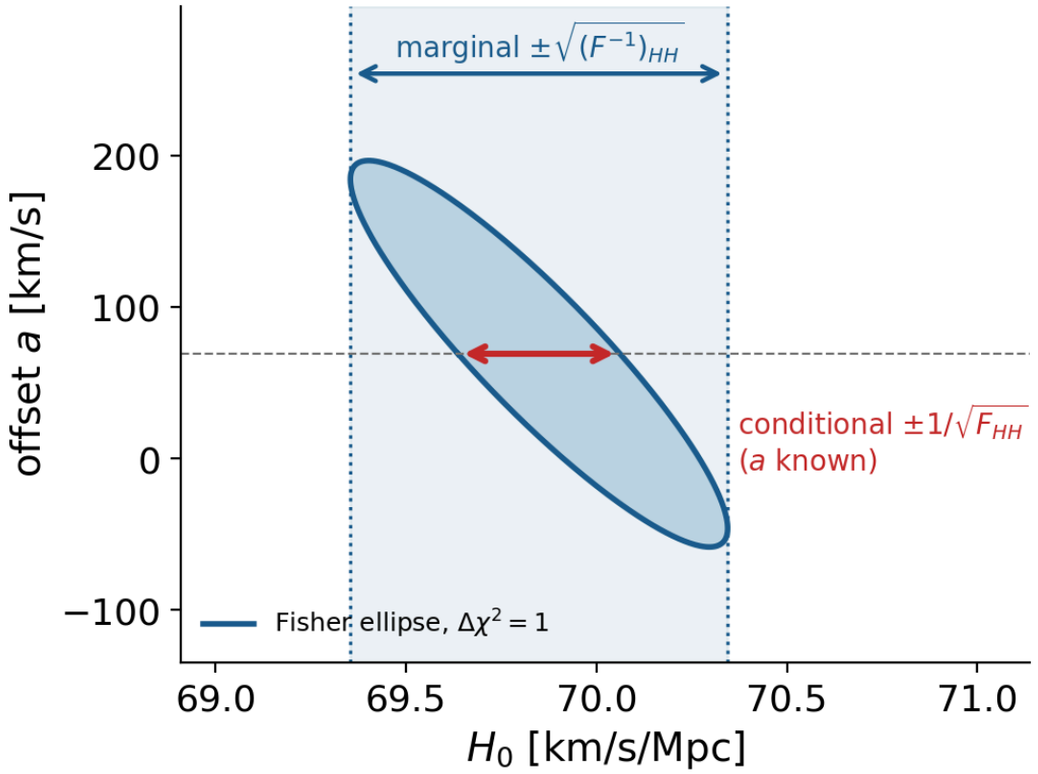


Now: the evidence, more samplers, Fisher forecasts for large-scale structure, live codes.

Fisher matrix: forecasting before the data

$$F_{ij} = \langle -\partial_i \partial_j \ln L \rangle = \sum_k \frac{1}{\sigma_k^2} \frac{\partial f_k}{\partial \theta_i} \frac{\partial f_k}{\partial \theta_j}$$

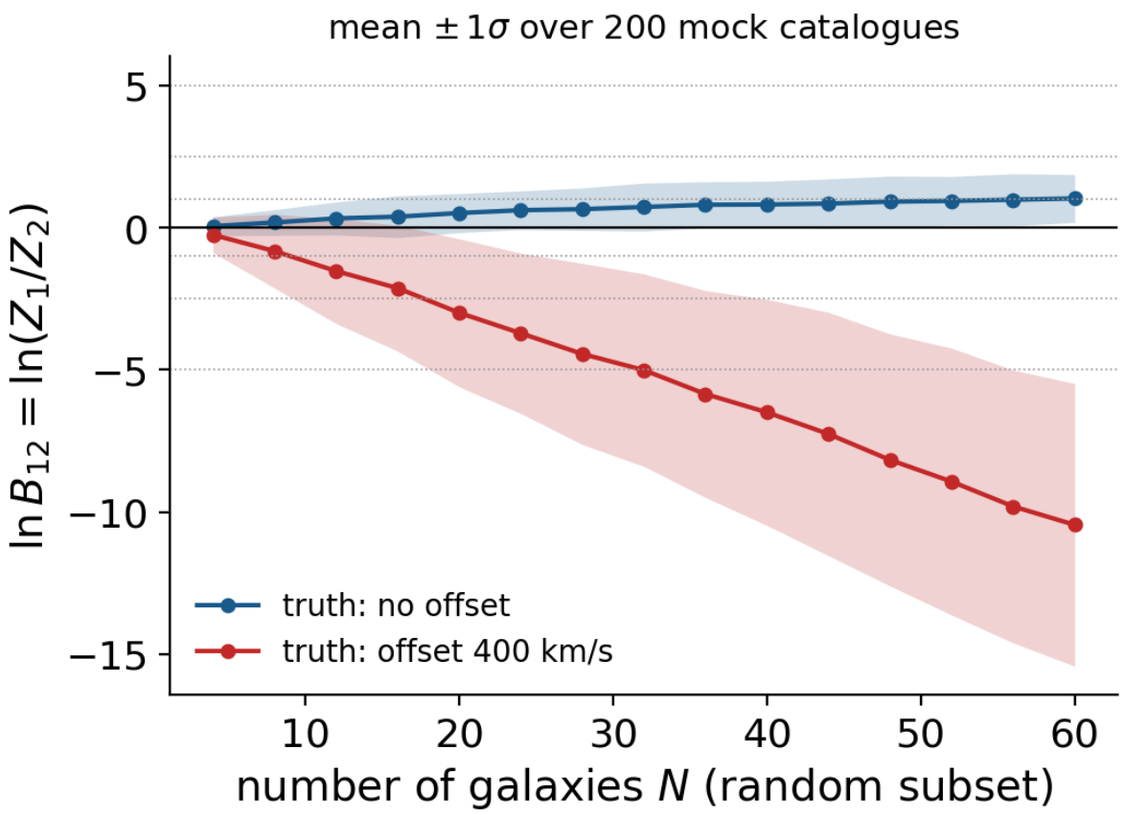
$$\sigma(\theta_i) \;\geq\; \sqrt{(F^{-1})_{ii}}$$



- Cramér–Rao: no unbiased estimator does better
- The marginal error is the one to quote; the conditional error $1/\sqrt{F_{ii}}$ assumes the other parameters are known
- Here: freeing the offset a more than doubles the error on H_0 (0.50 instead of 0.21 km/s/Mpc)
- Independent experiments: add the F matrices

Model selection: the evidence and Occam's razor

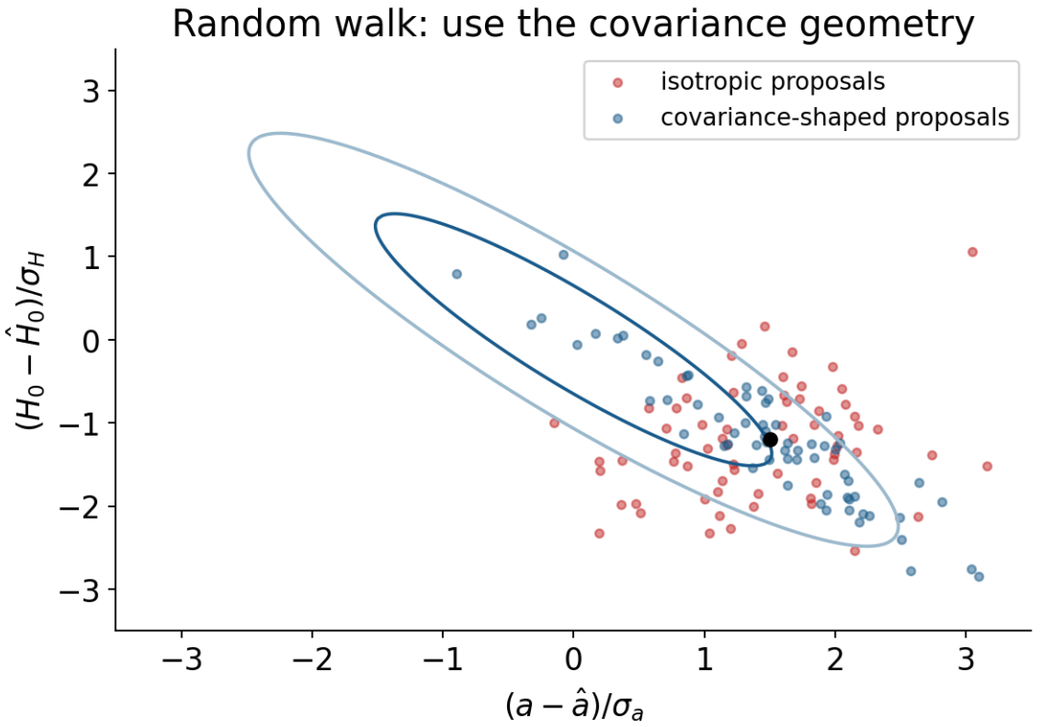
$$B_{12} = \frac{Z_1}{Z_2} = \frac{\int \mathrm{d}\boldsymbol{\theta}_1\, p(D|\boldsymbol{\theta}_1, M_1)\, p(\boldsymbol{\theta}_1|M_1)}{\int \mathrm{d}\boldsymbol{\theta}_2\, p(D|\boldsymbol{\theta}_2, M_2)\, p(\boldsymbol{\theta}_2|M_2)}$$



- M_1: $cz = H_0 d$ vs M_2: $cz = a + H_0 d$
- χ^2 never favours the simpler model; the evidence can, because extra prior volume is penalised
- Jeffreys scale on $|\ln B|$: below 1 inconclusive, 1–2.5 weak, 2.5–5 moderate, above 5 strong (Trotta)
- Curves: mean ± 1σ over 200 mock catalogues of up to $N = 60$ galaxies
- Truth $a = 0$ (M_1 true): $\ln B_{12} = 1.0 \pm 0.8$ at $N = 60$, a mild Occam preference for the simpler M_1
- Truth $a = 400$ km/s (M_2 true): $\ln B_{12} \approx -10 \pm 5$, strong preference for M_2, growing with N
- The result depends on the prior ranges (here $a \in [-500, 500]$)

Proposal geometry decides the efficiency

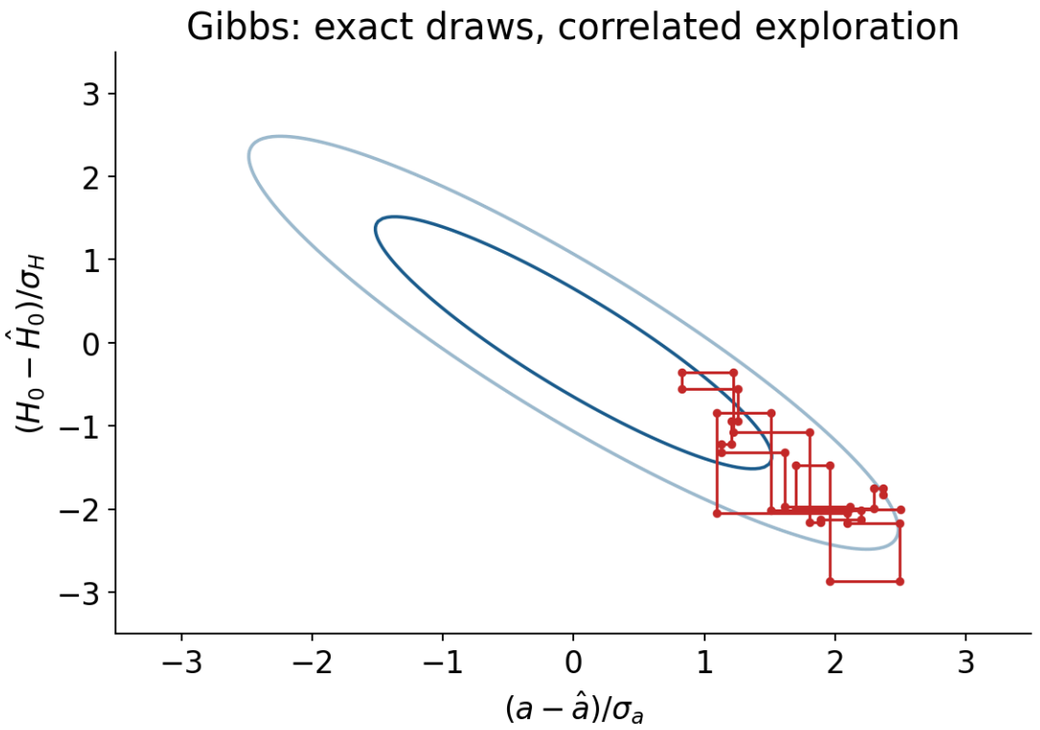
- An isotropic step must be as small as the **narrowest** direction, so a correlated posterior is explored slowly along its long axis
- For a Gaussian target in d dimensions the best random-walk step is about $2.38/\sqrt{d}$ times the posterior width; acceptance then tends to 23%
- Better: draw steps from the posterior covariance, tuned by an exploratory run (Padilla et al.) or taken from the Fisher matrix
- Reparametrising to less correlated parameters helps just as much



$$\boldsymbol{\theta}_c = \boldsymbol{\theta}_i + \mathcal{N}\left(0,\ \frac{2.38^2}{d}\,\mathbf{C}\right), \qquad \mathbf{C} \approx F^{-1}$$

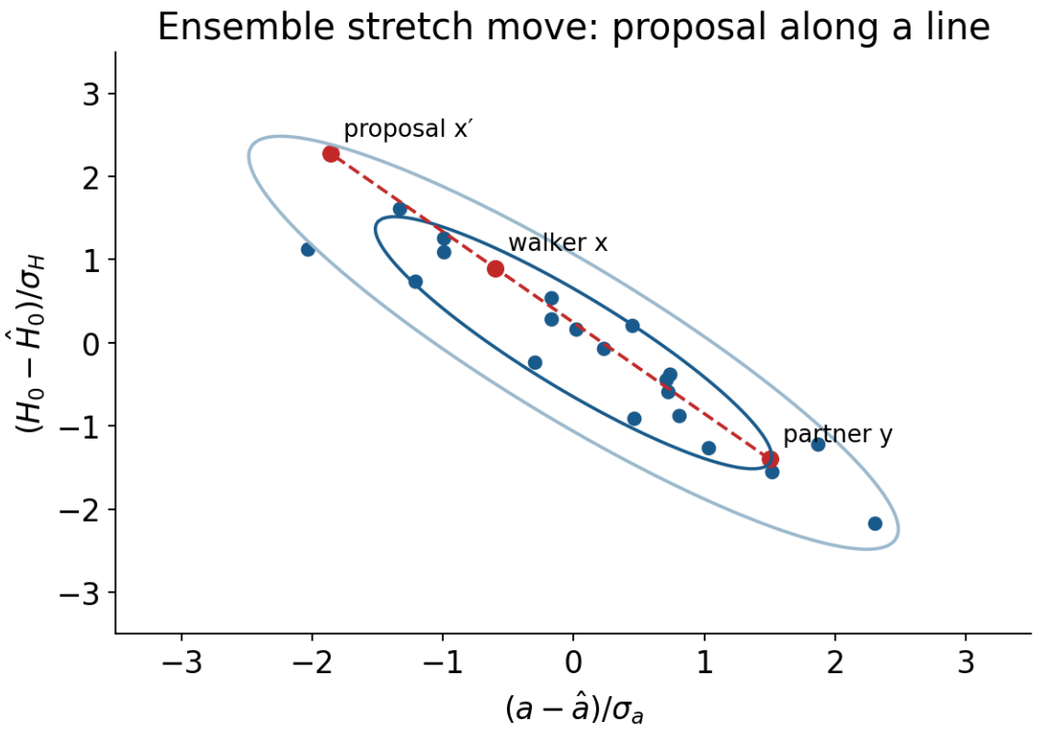
Gibbs sampling: one parameter at a time

- Draw each parameter from its **full conditional**: $\theta_1 \sim p(\theta_1|\theta_2, \dots)$, then $\theta_2 \sim p(\theta_2|\theta_1, \dots)$
- Special case of Metropolis–Hastings in which every move is accepted; no tuning
- Needs conditionals you can sample (conjugate priors, hierarchical models); in the Hubble toy both $p(a|H_0)$ and $p(H_0|a)$ are Gaussian
- Strong correlations give the staircase path in the figure: many small steps
- Used for CMB map-level inference and hierarchical models



Ensemble samplers: the stretch move

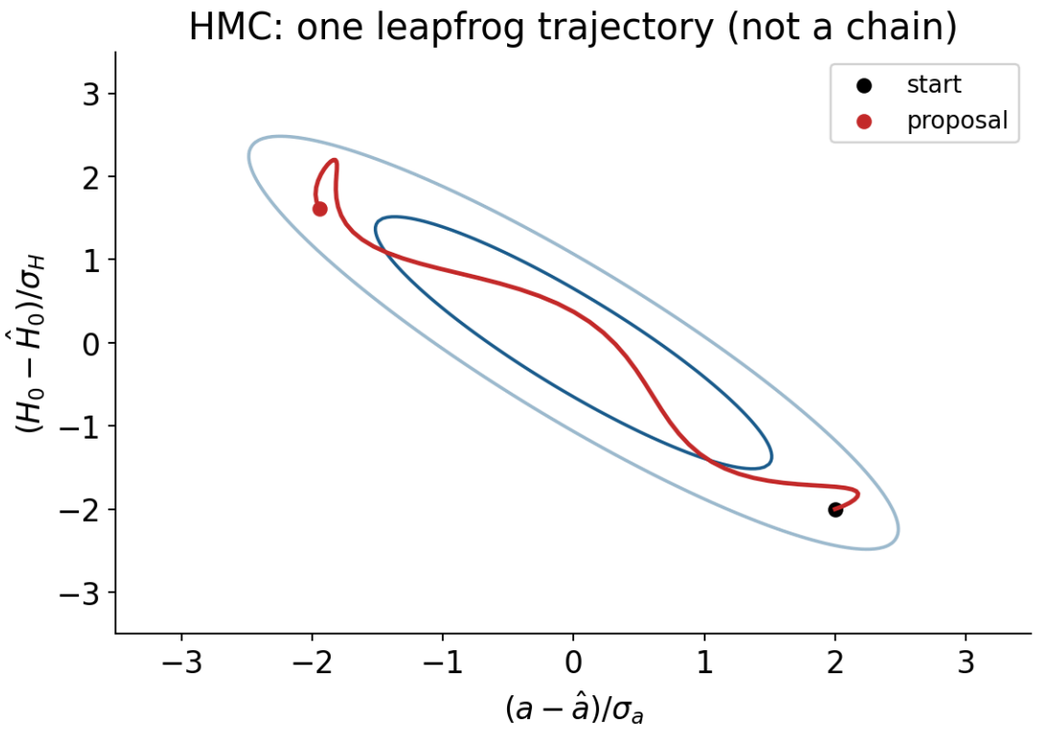
- Many ‘walkers’ move together; each is updated using a partner chosen from the others
- Affine invariance: a linear change of parameters, such as a correlated Gaussian, makes no difference, so no proposal covariance is needed
- Rule of thumb: a few times d walkers; **emcee** is the standard code
- Weak spots: very high dimension (tens of parameters) and multimodal posteriors
- Needs only the log-posterior function



$$\boldsymbol{\theta}' = \mathbf{y} + z(\boldsymbol{\theta} - \mathbf{y}), \quad g(z) \propto z^{-1/2}, \quad z \in \left[\tfrac{1}{a}, a\right]; \qquad p_{\mathrm{acc}} = \min\left[1,\, z^{d-1}\frac{p(\boldsymbol{\theta}')}{p(\boldsymbol{\theta})}\right]$$

Hamiltonian Monte Carlo

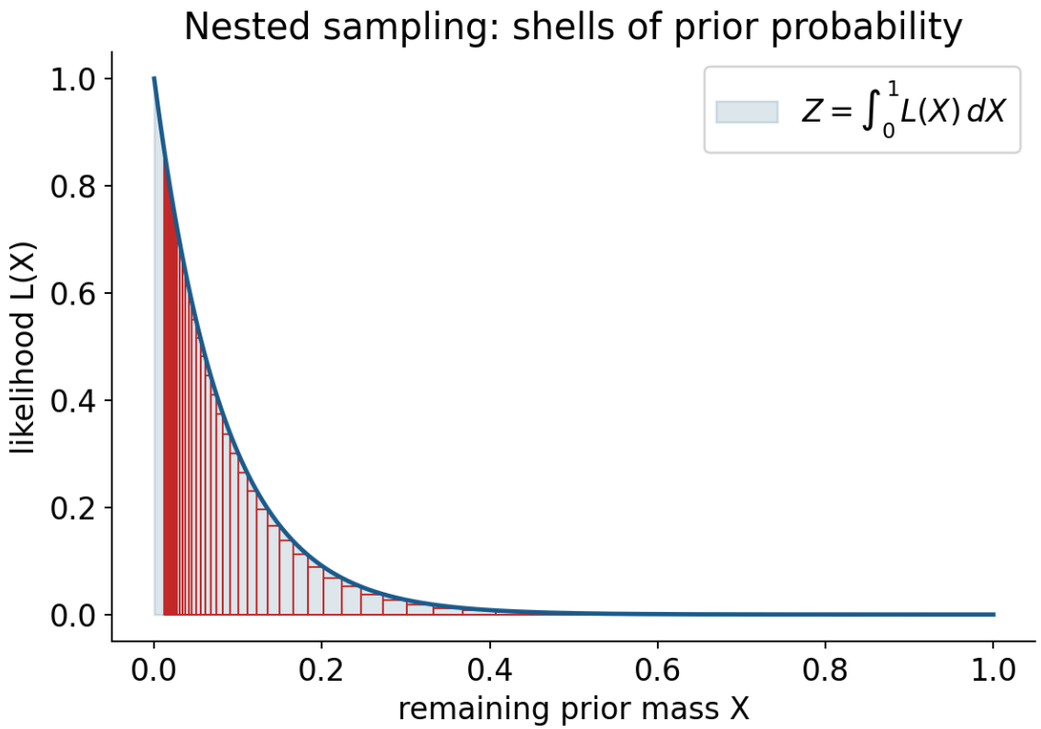
- Treat $U = -\ln p$ as a potential energy, add a momentum $\mathbf{q}$, and follow the Hamiltonian flow (leapfrog steps)
- Long, informed jumps: high acceptance even in many dimensions
- Costs: the gradient of ln p, and tuning of step size and trajectory length (the NUTS variant tunes the length automatically)
- Gradients mean autodiff: cloelib is built on JAX, but a full Boltzmann-solver call is not differentiable; emulators are
- Play with it: QR code, right



$$H(\boldsymbol{\theta}, \mathbf{q}) = U(\boldsymbol{\theta}) + \tfrac{1}{2}\mathbf{q}^T M^{-1}\mathbf{q}, \qquad p_{\text{acc}} = \min\left[1,\, e^{-\Delta H}\right]$$

Nested sampling: the evidence as a sum

- Reorder the prior volume by likelihood: $X(\lambda)$ is the prior mass with $L > \lambda$, so the evidence is a 1D integral
- N_{live} points; at each step discard the worst and replace it by a new point with higher L; volume shrinks as $X_i \approx e^{-i/N_{\text{live}}}$
- Posterior samples come as a by-product (weights $L_i \, \Delta X_i$)
- Handles multimodal posteriors; MultiNest, dynesty, **Nautilus**
- Run once for each model: this is how Bayes factors are computed



$$Z = \int L \, \mathrm{d}X \;\approx\; \sum_i L_i \, \Delta X_i$$

Which sampler for which problem?

Sampler	Needs	Evidence	Good for	Typical codes
Metropolis–Hastings	log-posterior; proposal covariance	no (extra step)	ΛCDM-type problems with a good covariance	MontePython, Cobaya, CosmoMC
Gibbs	sampleable conditionals	no	hierarchical models	custom
Ensemble (stretch)	log-posterior	no	moderate d, no tuning	emcee
HMC / NUTS	gradient of log-posterior	no	high d, smooth posteriors	numpyro, blackjax
Nested	log-likelihood, prior transform	yes	multimodal; model comparison	MultiNest, dynesty, Nautilus
Tempering (MC^3)	ladder of temperatures	possible	separated modes	custom, some codes

Practical rule. Use the simplest sampler that converges, check it on a case with a known answer (Hour 1), and compare with the Fisher forecast: when they disagree, learn why.

Fisher toolbox: combine, fix, marginalise

- **Combine** independent probes: $F = F_A + F_B$, parameters matched by name
- **Prior**: a Gaussian prior of width σ_p adds $1/\sigma_p^2$ to F_{ii}
- **Fix** a parameter: delete its row and column (conditional errors)
- **Marginalise**: invert F, keep the sub-block; $\sigma_i = \sqrt{(F^{-1})_{ii}}$
- **Reparametrise**: $F_\phi = J^T F_\theta J$, $J_{ij} = \partial\theta_i/\partial\phi_j$
- **Figure of merit** for (w_0, w_a): $\mathrm{FoM} = \sqrt{\det \tilde{F}_{w_0 w_a}}$, $\tilde{F}$ marginalised

In cosmicfishpie

- `fish_a + fish_b`
- `marginalise(F, names)`
- `reshuffle(F, names)`: others fixed
- `eliminate_parameters(F, names)`
- prior: add a 1×1 Fisher of $1/\sigma^2$ for that parameter

Toy model, Hour 1: freeing the offset a more than doubles $\sigma(H_0)$: conditional 0.21 vs marginal 0.50 km/s/Mpc. The same two numbers appear for every pair of correlated cosmological parameters; quoting the conditional error is the common mistake.

Fisher matrix for large-scale structure

Photometric (WL, GCph, GGL): sum over multipoles, C_ℓ is the matrix of all auxiliary tomographic spectra

$$F_{\alpha\beta} = \tfrac{1}{2} f_{\rm sky} \sum_\ell (2\ell+1)\, \mathrm{Tr}\left[C_\ell^{-1}\, \partial_\alpha C_\ell\, C_\ell^{-1}\, \partial_\beta C_\ell\right]$$

Spectroscopic (GCsp): sum over redshift bins i, integral over k, μ; V_i is the bin volume

$$F_{\alpha\beta} = \sum_i \tfrac{1}{8\pi^2} \int \mathrm{d}\mu \int k^2 \mathrm{d}k\; \partial_\alpha \ln P_{\rm obs}\, \partial_\beta \ln P_{\rm obs}\; V_i$$

Photometric ingredients

- 10 equi-populated bins, z up to 2.5; 30 galaxies/arcmin²; shape noise 0.3
- $f_{\rm sky} = 0.3636$; $\ell_{\rm max}$ pessimistic 1500 (WL), 750 (GCph); optimistic 5000, 1500
- Nuisance: 10 galaxy biases, 2 intrinsic-alignment amplitudes

Spectroscopic ingredients

- 4 bins, 0.9 < z < 1.8; volumes 8–16 Gpc³/h³
- $k_{\rm max}$ = 0.25 (pess.), 0.30 (opt.) h/Mpc
- $P_{\rm obs}$: bias, RSD, Alcock–Paczyński, shot noise, redshift errors
- Nuisance: $\ln[b\sigma_8]$ and P_s in each bin

Cosmological parameters: Ω_b, Ω_m, h, n_s, σ_8, w_0, w_a. Source: Casas et al., arXiv:2303.09451, eqs. 8 and 10, tables 1–3.

Derivatives: step sizes matter

- CosmicFish: first derivatives of the observables, step 1% of the fiducial value (2N evaluations); MontePython/Fisher: second derivatives of ln L (2N² evaluations), step $x\,\sigma_{\rm marg}$ with $x = 0.1$ (photo) or 0.05 (spectro)
- Too small a step: solver noise. Too large: non-Gaussian terms, $F(x) \propto x^2$ (figure)
- **Fisher needs more solver accuracy than MCMC**: CAMB Halofit tolerance 10^{-6} instead of 10^{-3}, CLASS high precision; with default precision errors are off by up to 11% or more
- Neutrino paper: 10% step for Σm_ν; 1% is dominated by solver noise

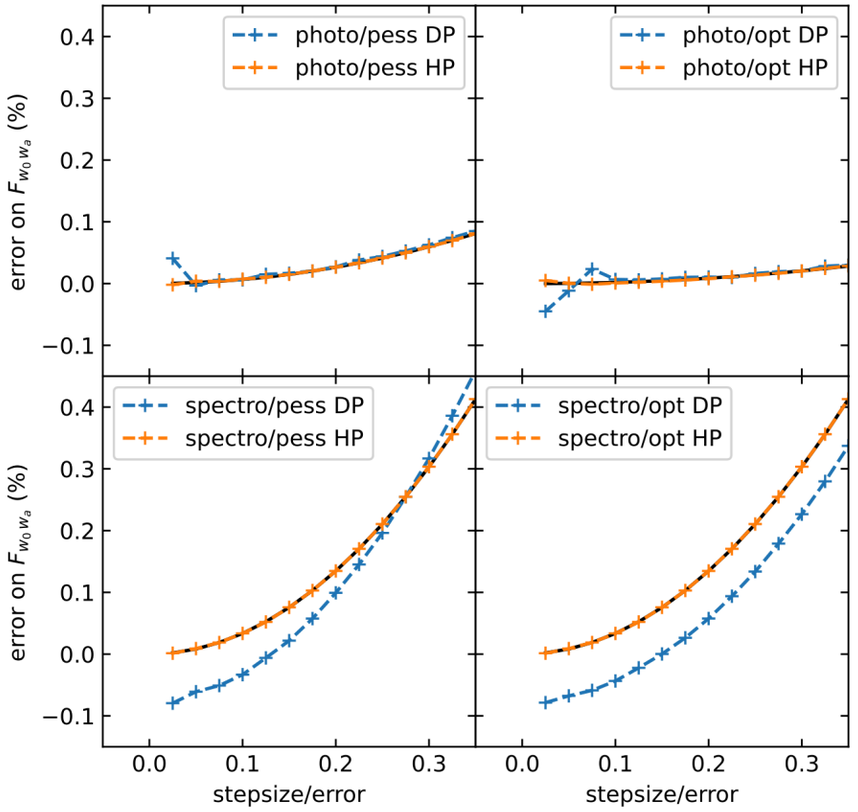


Fig. 19, Casas et al., arXiv:2303.09451

Validation against MCMC

- Five Fisher pipelines (CosmicFish with CAMB or CLASS, internal or external spectra, MontePython/Fisher) plus a MontePython MCMC; ΛCDM + w_0w_a
- All within 10% of the median; CosmicFish vs MP/Fisher ≲ 2% (pessimistic); CAMB vs CLASS below 0.5%
- Photometric contours are almost elliptical: Fisher ≈ MCMC. Spectroscopic: slight non-Gaussianity, Fisher still excellent

1σ, photo pess.	CosmicFish	MP/Fisher	MP/MCMC
n_s	0.010	0.010	0.011
σ_8	0.0041	0.0042	0.0044
w_0	0.043	0.044	0.045
w_a	0.17	0.18	0.18

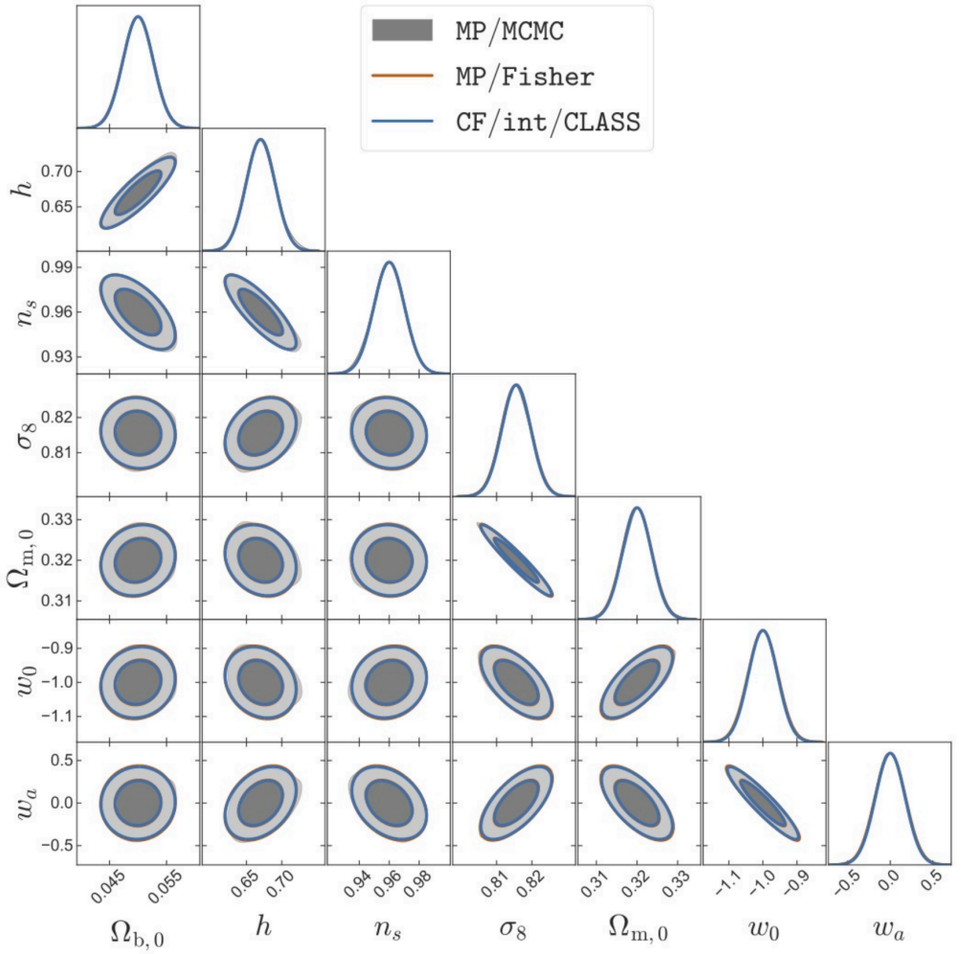
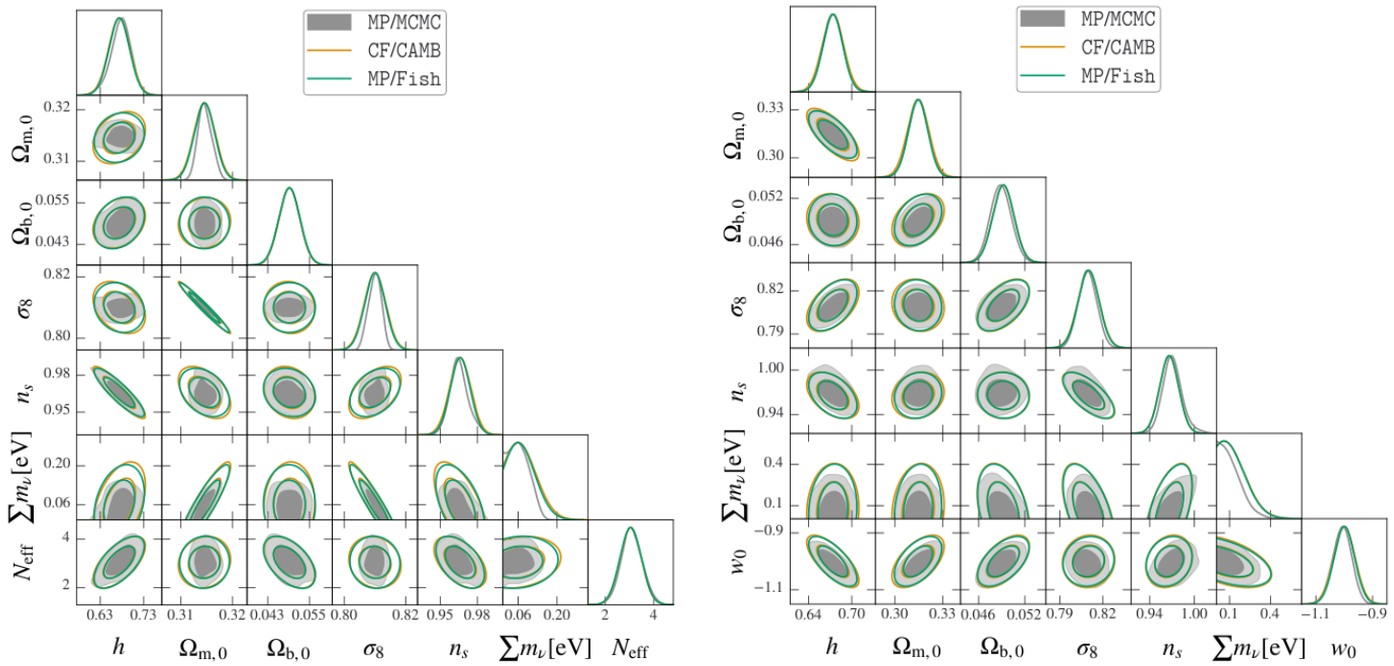


Fig. 4, Casas et al., arXiv:2303.09451 (grey: MP/MCMC)

Where Fisher fails: prior boundaries



- $\Sigma m_\nu > 0$ truncates the posterior: a Gaussian cannot
- The cut propagates to h, Ω_m, σ_8, n_s; MCMC contours are tighter than the Fisher ellipses
- 'Banana' in n_s–Σm_ν with w_0 free
- The full 9-parameter model: Fisher unstable (step sizes), so the paper uses MCMC

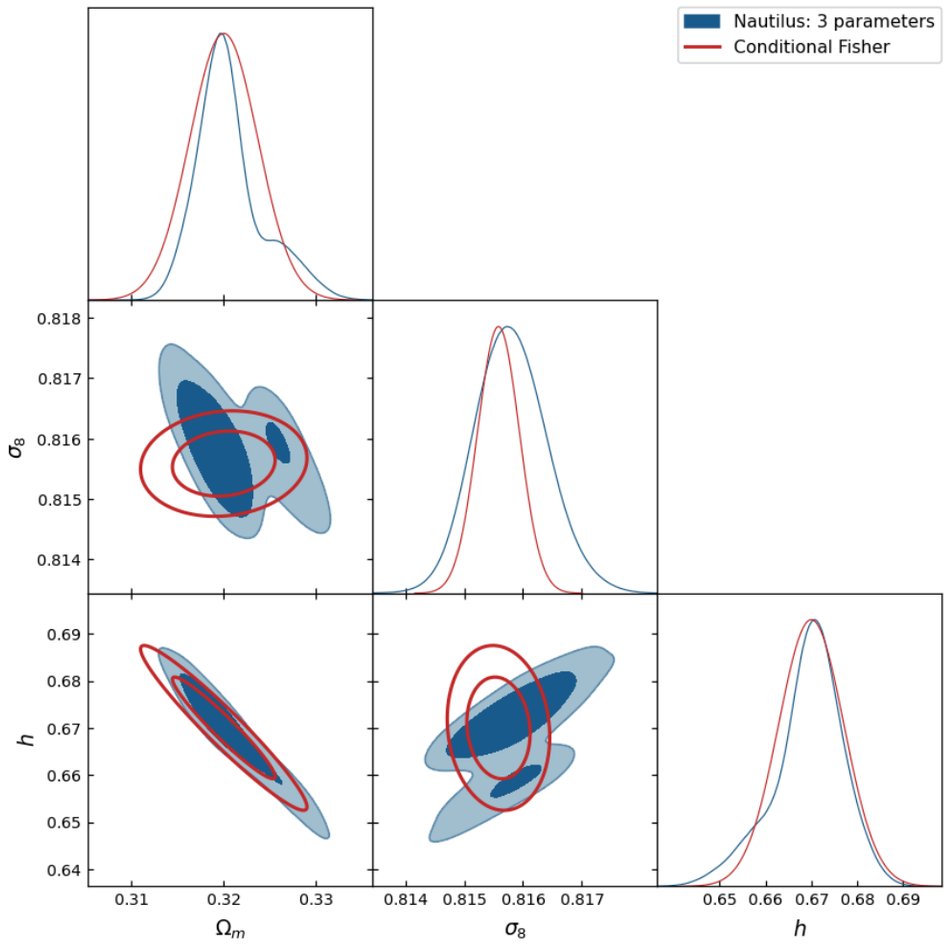
Left: ΛCDM + Σm_ν + N_{eff}; right: w_0CDM + Σm_ν (spectroscopic). Figs. 10–11, Euclid Collaboration, arXiv:2405.06047 (grey: MCMC).

Result (MCMC, pessimistic, ΛCDM + Σm_ν)

- $\sigma(\Sigma m_\nu)$ = 56 meV Euclid alone; 23 meV with Planck; 16 meV with CMB-S4 + LiteBIRD
- Euclid + Planck: 2.6σ for non-zero mass; with CMB-S4 + LiteBIRD almost 4σ

Our own check: Nautilus vs Fisher

- cosmicfishpie GCsp, symbolic spectra; sample (Ω_m, σ_8, h) with Nautilus, nuisance parameters fixed
- Matching reference is the **conditional** Fisher (`reshuffle`), not the marginalised one
- Rehearsal: 409 s on 8 workers (300 live points, ESS ≈ 1200); the instructor job on freyja runs beforehand
- **Ω_m and h agree**: σ = 0.0037 vs 0.0037 and 0.0078 vs 0.0072
- **σ_8 does not**: sampled σ = 0.00061, Fisher 0.00036; correlations differ



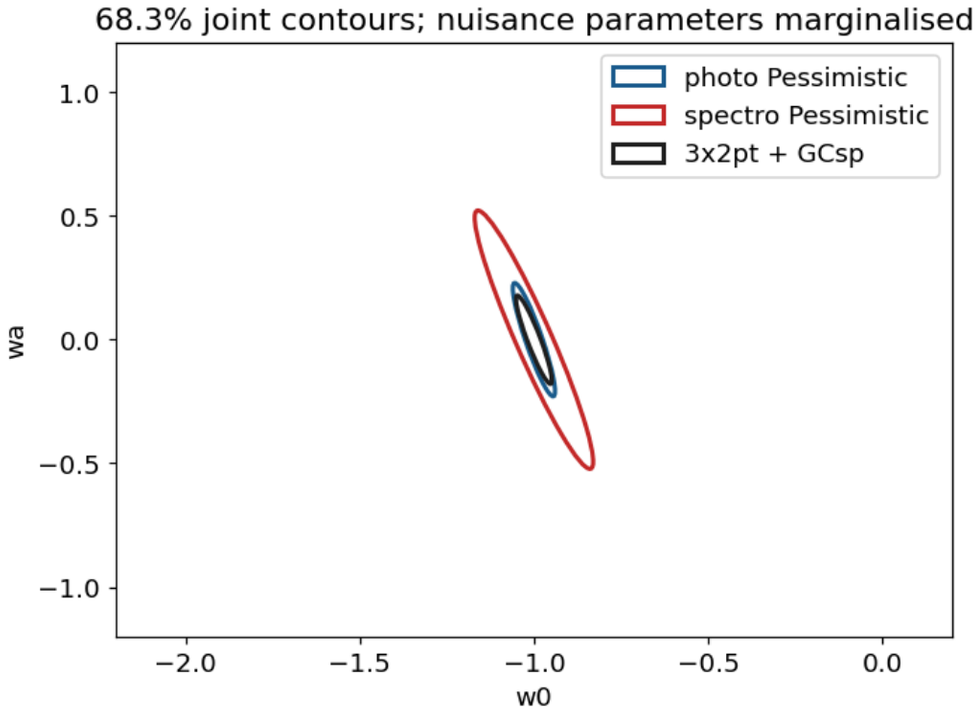
A finite-difference Hessian of the same ln L supports the sampler.
Open question: follow up before quoting.

Live: cosmicfishpie forecast

Notebook `cfp_live_demo.ipynb` (kernel 'lecture')

1. Symbolic w_0w_aCDM forecast: 3×2pt in 7.8 s, GCsp in 2.7 s
2. Add them: `F_photo + F_spectro`
3. Marginalise vs reshuffle: $\sigma(w_0)$ 0.040 vs 0.0076
4. Add a prior on h ($\sigma = 0.01$): FoM 391 → 509
5. Optimistic vs pessimistic: FoM 391 → 1784

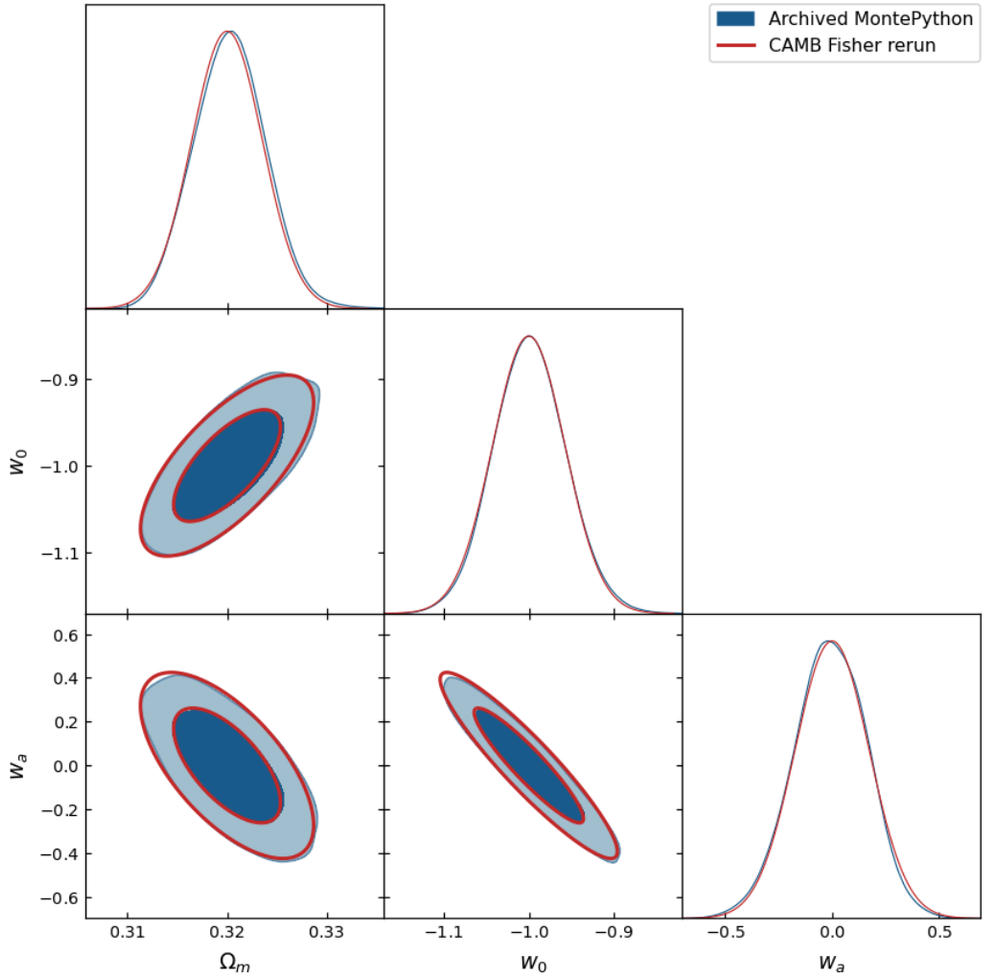
Caveat. The symbolic backend is fast but approximate: errors differ from CLASS by up to 14% (photo; 32% on n_s) and 10–33% (spectro). CAMB and CLASS agree to better than 1%.



68% contours, w_0–w_a: photo (blue), spectro (red), combined (black)

FoM (w_0, w_a): photo 391, GCsp 78, combined 647

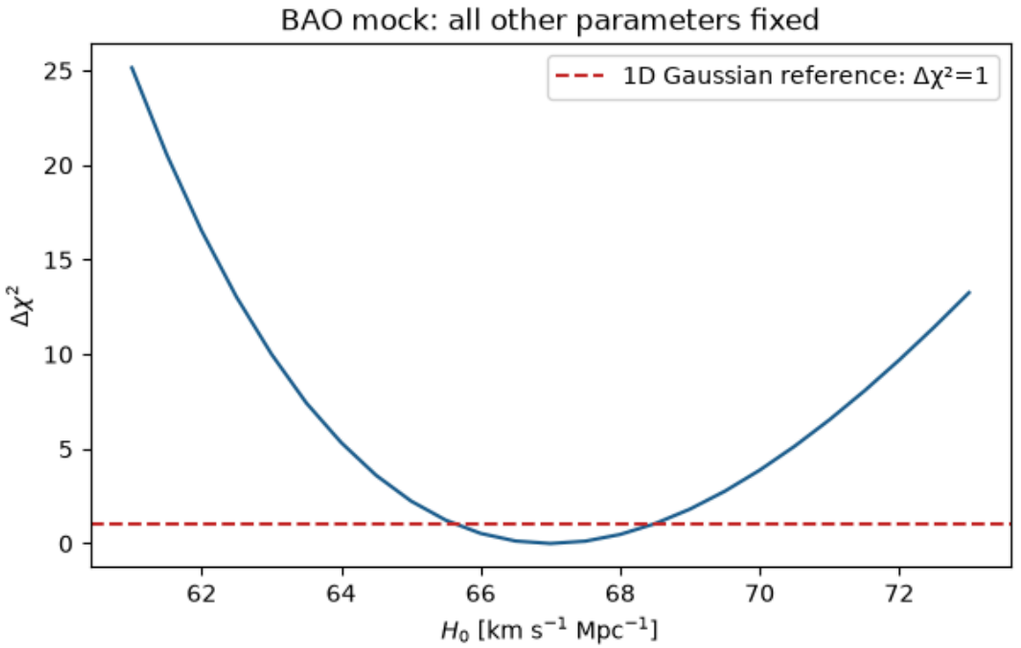
Live: archived MontePython vs Fisher



- Archived MontePython chain (ΛCDM + w_0w_a, photometric pessimistic) against a Gaussian from a CAMB Fisher
- The Fisher matrix is a **modern rerun** of the baseline setup with CAMB and CLASS (they agree below 0.5%), not the published matrix
- Ω_m, w_0, w_a: the Gaussian traces the chain
- Spectroscopic reruns come out 30–40% wider than the published CosmicFish numbers (different nuisance set); we compare only the photometric case

cloelib and cloelike: a loglike function

- **cloelib**: theory (background, perturbations, observables), JAX-based; **cloelike**: Euclid likelihoods with `loglike(parameters)`
- Demo: spectroscopic BAO likelihood with public Euclid data (4 redshift bins, Zenodo)
- $\ln L$ at $H_0 = 70$: −1.938, equal to the repository regression value; setup 3–6 s, 25-point H_0 scan 0.6–1.5 s
- Own environment (Python 3.12, numpy 2) because cosmicfishpie needs numpy < 2
- No built-in Cobaya or MontePython wrapper: pass `loglike` to Nautilus, dynesty or emcee yourself




```
like = EuclidLikelihood_GCspectro_BAO(data=data, Background=CAMBBackground,
LinearPerturbations=CAMBLinearPerturbations)
like.loglike({'H0': 70, 'Omega_cdm0': 0.27, 'Omega_b0': 0.049, ...})
```

Exercises and take-home

Exercises (notebooks)

1. Add a prior σ(h) = 0.01 in `cfp_live_demo`. Which parameters gain, and how does the FoM change?
2. Compare `marginalise` and `reshuffle` for (w_0, w_a). Which is the error you would quote?
3. Pessimistic → optimistic: which probe gains most?
4. Random-walk Metropolis on a correlated Gaussian in $d = 2, 10, 50$: measure the acceptance for steps $2.38/\sqrt{d}$ and compare with an isotropic guess.
5. Bonus: explain the σ_8 mismatch between Nautilus and the Fisher matrix.

Take-home

- Covariance-tuned Metropolis is the workhorse; ensemble samplers need no tuning; HMC needs gradients; nested sampling gives the evidence
- Fisher: add, fix, marginalise, reparametrise; quote marginal errors
- A Fisher forecast is only as good as its derivatives: step size and solver precision
- It fails at prior boundaries and for non-Gaussian posteriors: check with MCMC
- A likelihood is a function: cloelike plugs into any sampler
- Check convergence; state your priors

References

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- L. Verde, Statistical methods in cosmology, arXiv:0911.3105
- A. Heavens, Statistical techniques in cosmology, arXiv:0906.0664
- R. Trotta, Bayes in the sky, arXiv:0803.4089; Bayesian methods in cosmology, arXiv:1701.01467
- Euclid Collaboration: S. Casas et al., Validation of the MontePython forecasting tools, arXiv:2303.09451
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- G. O. Roberts, A. Gelman, W. R. Gilks, Weak convergence and optimal scaling of random walk Metropolis algorithms, Ann. Appl. Probab. 7 (1997)
- J. Goodman, J. Weare, Ensemble samplers with affine invariance, Commun. Appl. Math. Comput. Sci. 5 (2010); D. Foreman-Mackey et al., emcee, arXiv:1202.3665
- R. M. Neal, MCMC using Hamiltonian dynamics, arXiv:1206.1901
- J. Skilling, Nested sampling for general Bayesian computation, Bayesian Analysis 1 (2006)
- M. Bonici et al., cloelib, arXiv:2605.23839; cloelike, arXiv:2605.23841
- Interactive MCMC gallery: chi-feng.github.io/mcmc-demo

Backup: from the toy model to ΛCDM

Replace the straight line by H(z) and the mock data by real measurements. Flat ΛCDM, no radiation (Padilla et al.):

$$H^2(z) = H_0^2 \left[\Omega_m (1+z)^3 + 1 - \Omega_m\right]$$

Parameters (H_0, Ω_m).

Cosmic chronometers only (flat priors)

- $H_0 = 67.8 \pm 3.1$ km/s/Mpc
- $\Omega_m = 0.33 \pm 0.06$

BAO + CMB (compressed) + SNe Ia + chronometers

- $H_0 = 68.4 \pm 0.6$ km/s/Mpc
- $\Omega_m = 0.299 \pm 0.007$ (from $h = 0.684 \pm 0.006$)

Adding data shrinks the posterior; extending the model (w, $w_0 w_a$, curvature) lowers χ^2_{min} but costs evidence.

Same structure as Part II: model → χ^2 → priors → sampler → contours. Only the theory call is more expensive.

Backup: anatomy of an analysis



Toy → real

- Line a + H_0d → angular power spectra, correlation functions, BAO
- 300 km/s errors → full covariance matrix

Fisher side

- Derivatives of t(θ) and a covariance give F_{ij} with no sampling: cosmicfishpie

cloelike only evaluates the Gaussian χ² given data, covariance and theory predictions; sampling is left to external tools.